

KIMBALL, NE, USA

WMO: 725665

Lat: 41.189N

Lon: 103.671W

Elev: 4894

StdP: 12.28

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 94073

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | -3.0       | 3.0 | -8.9                          | 4.1 | -0.7 | -3.7 | 5.4 | 7.5  | 36.8                  | 38.6 | 33.8 | 30.6 | 8.8                   | 250  | 0.666 |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 28.8                   | 94.2            | 60.7 | 90.8 | 60.6 | 88.1 | 60.5 | 66.4                | 79.8 | 65.1 | 79.0 | 63.9 | 78.3 | 10.2                 | 150  |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 63.2                            | 104.2 | 68.9 | 61.4 | 97.7 | 69.0 | 59.6 | 91.4 | 67.9 | 34.0          | 79.0 | 32.9 | 79.1 | 31.8 | 77.3 | 72.3              |

Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |       |                    | n-Year Return Period Values of Extreme Temperature |           |       |            |       |            |       |            |       |       |
|-------------------|------|------|----------------------------|------|-------|--------------------|----------------------------------------------------|-----------|-------|------------|-------|------------|-------|------------|-------|-------|
| 1%                |      | 2.5% | 5%                         | Mean |       | Standard Deviation |                                                    | n=5 years |       | n=10 years |       | n=20 years |       | n=50 years |       |       |
| Min               | Max  | Min  | Max                        | Min  | Max   | Min                | Max                                                | Min       | Max   | Min        | Max   | Min        | Max   | Min        | Max   |       |
| (a)               | (b)  | (c)  | DB                         |      | (e)   | (f)                | (g)                                                | (h)       | (i)   | (j)        | (k)   | (l)        | (m)   | (n)        | (o)   | (p)   |
| 31.7              | 26.8 | 23.8 | WB                         |      | -13.3 | 99.1               | 5.5                                                | 2.5       | -17.2 | 100.9      | -20.4 | 102.4      | -23.5 | 103.8      | -27.5 | 105.6 |
|                   |      |      |                            |      | -13.9 | 69.2               | 5.3                                                | 1.7       | -17.7 | 70.4       | -20.8 | 71.3       | -23.8 | 72.3       | -27.7 | 73.5  |

Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 48.5   | 29.2  | 29.4  | 39.6  | 45.2  | 54.0  | 65.4  | 72.7  | 69.6  | 61.6  | 48.1  | 38.1  | 28.1  |
|                                                                   | DBStd       | 17.93  | 10.79 | 11.68 | 9.84  | 8.58  | 8.79  | 6.86  | 5.26  | 5.34  | 7.78  | 9.49  | 10.29 | 11.33 |
|                                                                   | HDD50       | 3012   | 645   | 578   | 337   | 187   | 61    | 1     | 0     | 0     | 10    | 145   | 368   | 680   |
|                                                                   | HDD65       | 6574   | 1110  | 997   | 786   | 595   | 350   | 78    | 9     | 23    | 152   | 523   | 807   | 1144  |
|                                                                   | CDD50       | 2476   | 0     | 1     | 16    | 42    | 186   | 463   | 705   | 609   | 356   | 87    | 11    | 0     |
|                                                                   | CDD65       | 565    | 0     | 0     | 0     | 0     | 10    | 89    | 249   | 167   | 49    | 1     | 0     | 0     |
|                                                                   | CDH74       | 8148   | 0     | 0     | 12    | 43    | 304   | 1403  | 3141  | 2135  | 974   | 133   | 3     | 0     |
|                                                                   | CDH80       | 3494   | 0     | 0     | 0     | 2     | 68    | 570   | 1565  | 912   | 359   | 18    | 0     | 0     |
|                                                                   | WSAvg       | 10.7   | 12.0  | 11.5  | 12.0  | 12.3  | 10.8  | 9.9   | 8.9   | 8.8   | 9.4   | 10.6  | 10.9  | 11.7  |
|                                                                   | PrecAvg     | 16.9   | 0.4   | 0.3   | 1.1   | 1.6   | 2.8   | 2.8   | 2.8   | 1.8   | 1.4   | 1.0   | 0.6   | 0.4   |
| Precipitation                                                     | PrecMax     | 25.0   | 1.3   | 0.9   | 2.8   | 3.9   | 7.9   | 6.7   | 7.3   | 4.8   | 4.7   | 3.1   | 1.5   | 1.3   |
|                                                                   | PrecMin     | 6.7    | 0.0   | 0.0   | 0.1   | 0.1   | 0.6   | 0.8   | 0.4   | 0.0   | 0.1   | 0.0   | 0.0   | 0.0   |
|                                                                   | PrecStd     | 3.8    | 0.3   | 0.2   | 0.8   | 0.9   | 1.6   | 1.6   | 1.7   | 1.1   | 1.1   | 0.8   | 0.4   | 0.4   |
|                                                                   |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 62.6  | 66.5  | 75.6  | 79.0  | 86.4  | 96.6  | 99.0  | 95.1  | 92.3  | 82.3  | 73.2  | 61.0  |
|                                                                   |             | MCWB   | 42.3  | 45.6  | 49.4  | 50.1  | 56.6  | 59.5  | 61.2  | 59.8  | 59.9  | 52.9  | 48.2  | 42.0  |
|                                                                   | 2%          | DB     | 54.7  | 58.8  | 69.5  | 73.6  | 81.4  | 90.7  | 95.2  | 91.2  | 88.1  | 78.2  | 67.6  | 54.7  |
|                                                                   |             | MCWB   | 39.1  | 41.5  | 46.3  | 49.7  | 55.2  | 59.9  | 61.5  | 60.4  | 58.2  | 52.3  | 46.7  | 39.6  |
|                                                                   | 5%          | DB     | 49.6  | 52.3  | 64.1  | 69.0  | 77.3  | 86.5  | 91.8  | 88.9  | 84.4  | 72.9  | 61.5  | 49.7  |
|                                                                   |             | MCWB   | 37.2  | 38.7  | 44.3  | 48.3  | 54.7  | 59.8  | 61.3  | 60.4  | 57.2  | 50.5  | 43.6  | 37.4  |
|                                                                   | 10%         | DB     | 44.4  | 45.9  | 58.8  | 63.6  | 72.3  | 82.1  | 89.5  | 85.8  | 80.6  | 67.2  | 55.3  | 44.8  |
|                                                                   |             | MCWB   | 35.0  | 35.8  | 42.5  | 46.4  | 53.3  | 59.1  | 61.4  | 60.8  | 56.1  | 49.0  | 40.9  | 35.0  |
|                                                                   | 0.4%        | WB     | 43.2  | 45.8  | 50.9  | 54.7  | 61.6  | 66.9  | 68.4  | 68.0  | 64.5  | 56.8  | 50.0  | 42.9  |
|                                                                   |             | MCDB   | 58.0  | 64.8  | 69.7  | 71.3  | 74.8  | 80.3  | 80.7  | 80.2  | 79.1  | 73.3  | 70.0  | 58.4  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 40.1  | 42.1  | 47.8  | 51.3  | 59.0  | 64.8  | 66.7  | 66.0  | 62.0  | 54.3  | 47.1  | 40.5  |
|                                                                   |             | MCDB   | 52.2  | 56.9  | 65.8  | 67.6  | 72.6  | 79.2  | 80.4  | 78.4  | 75.9  | 70.0  | 65.8  | 53.6  |
|                                                                   | 5%          | WB     | 37.7  | 39.1  | 45.4  | 49.4  | 56.9  | 63.0  | 65.6  | 64.4  | 60.1  | 52.1  | 44.2  | 37.8  |
|                                                                   |             | MCDB   | 47.8  | 51.0  | 62.3  | 66.1  | 71.1  | 78.2  | 79.6  | 77.4  | 74.1  | 68.4  | 60.0  | 48.3  |
|                                                                   | 10%         | WB     | 35.4  | 36.2  | 43.0  | 47.4  | 54.9  | 61.2  | 64.3  | 63.2  | 58.6  | 49.9  | 41.6  | 35.2  |
|                                                                   |             | MCDB   | 43.8  | 45.6  | 57.3  | 62.1  | 68.5  | 76.4  | 79.1  | 76.7  | 73.6  | 65.2  | 55.4  | 44.1  |
|                                                                   | 5% DB       | MDBR   | 22.6  | 23.0  | 26.1  | 25.3  | 25.6  | 27.7  | 28.8  | 28.1  | 29.1  | 27.0  | 25.7  | 22.7  |
|                                                                   |             | MCDBR  | 28.9  | 31.3  | 35.9  | 36.9  | 34.2  | 34.4  | 34.1  | 33.8  | 35.8  | 38.1  | 34.7  | 29.1  |
|                                                                   |             | MCWBR  | 16.5  | 17.0  | 17.6  | 16.2  | 13.1  | 11.7  | 9.2   | 9.6   | 12.3  | 16.5  | 17.2  | 16.7  |
|                                                                   |             | MCDBR  | 26.6  | 29.6  | 33.4  | 33.1  | 28.7  | 29.6  | 27.8  | 27.1  | 28.8  | 32.9  | 33.4  | 27.3  |
|                                                                   | 5% WB       | MCWBR  | 16.3  | 16.7  | 17.3  | 15.9  | 12.6  | 12.3  | 9.5   | 9.8   | 11.2  | 15.2  | 17.0  | 16.4  |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.239 | 0.247 | 0.266 | 0.300 | 0.321 | 0.325 | 0.353 | 0.345 | 0.307 | 0.272 | 0.253 | 0.240 |
|                                                                   | taud        |        | 2.608 | 2.582 | 2.551 | 2.450 | 2.464 | 2.479 | 2.433 | 2.443 | 2.533 | 2.602 | 2.576 | 2.578 |
|                                                                   | Ebn at Noon |        | 303   | 314   | 316   | 309   | 303   | 300   | 290   | 290   | 297   | 300   | 293   | 292   |
|                                                                   | Edh at Noon |        | 21    | 25    | 29    | 34    | 35    | 34    | 36    | 34    | 29    | 24    | 21    | 19    |
| All-Sky Solar Radiation                                           | RadAvg      |        | 747   | 1014  | 1413  | 1725  | 1982  | 2328  | 2283  | 2007  | 1634  | 1142  | 813   | 626   |
|                                                                   | RadStd      |        | 33    | 64    | 110   | 119   | 161   | 122   | 88    | 61    | 86    | 91    | 45    | 39    |

Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (19 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page